



Project Plan

Internship assignment:

Smart home integratie in Experience Lab



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Internship Company

The internship is carried out at Thomas More Kempen - Research Group Mobilab & Care. Mobilab & Care is an applied research group that focuses on smart living and care, with the goal of improving quality of life, safety, and independence in everyday life. The research group develops and evaluates practical solutions that can be used in real care and home environments, combining technology with a strong focus on feasibility, usability, and impact.

A key strength of Mobilab & Care is the use of a living-lab approach. Instead of only working with theoretical designs, ideas are tested in realistic settings and through concrete scenarios. This allows the team to validate whether a solution is not only technically possible, but also understandable, reliable, and acceptable for users and stakeholders. The Experience Lab environment supports experimentation with smart-home technologies and routine-based scenarios that reflect real life, such as night-time safety, assistance routines, and user-friendly interaction methods.

During the internship I work under the guidance of my work placement mentors Bert Bonroy and Glen Debard, and with Anke Coomans as my internship supervisor. Together we align on the scope and expected deliverables, and I receive feedback on feasibility, design choices, and documentation quality. The project is executed in the living-lab environment, where concepts can be translated into a demonstrable prototype and evaluated through scenario-based testing, logging, and iteration. The goal is to deliver not only a working prototype, but also clear documentation and evidence that supports future improvements and reuse of the work within Mobilab & Care.



Project Background

Many older adults prefer to stay in their own home as long as possible. Remaining at home supports autonomy and control (choosing when to sleep, eat, or receive visitors), identity and dignity (being in a familiar personal space), and continuity of routines (daily habits that reduce stress and confusion). It also helps maintain social and community connections, such as neighbours, local shops, and familiar routes, which can reduce feelings of isolation. For many people, home is strongly linked to psychological comfort and a sense of safety, which are important elements of quality of life.

At the same time, staying at home can increase exposure to safety risks, especially when physical capacity changes and the home environment is not adapted. The risks are often not linked to rare events; they happen during normal routines. One of the most common and risky routine moments is getting out of bed at night and walking to the bathroom. Low light, sleepiness, slower reaction time, dizziness, and reduced balance increase fall risk in this scenario.

Residential care facilities (sometimes called “care homes” or “nursing homes”) can provide structured support and supervision, and for some people they are the best or only safe option. However, moving to a facility can also be a major life transition. People may experience loss of independence, reduced privacy, separation from familiar environments, and disruption of routines. These factors can negatively affect perceived wellbeing for some older adults, especially when the move is not fully voluntary or happens during a period of health decline.

Because most homes do not have high-end domotics infrastructure, the internship assignment focuses first on a low-end retrofit approach: a solution that can be installed in almost any home with minimal installation effort, without rewiring and without requiring an existing domotics system. In this project, “low-end” means that the system relies on wireless sensors and wireless lighting controlled by a local hub (Raspberry Pi). The automation logic runs locally through Home Assistant. The system should provide safety support without invasive monitoring (no cameras) and without creating a feeling of being constantly watched.

The intention of this project is not to argue against residential care, but to support a realistic goal shared by many older adults and families: staying at home safely for longer when it is possible and desired, by reducing avoidable risks during high-risk routines and providing calm, low-effort support.



Objectives

The overall objective of this internship is to design, implement, and document a smart-home integration prototype in the Experience Lab that supports safety, comfort, and independence for older adults in a realistic home environment.

The first objective is to create a central local platform in which different devices and automations can work together. For this purpose, Home Assistant is used as the main control platform, running locally on a Raspberry Pi. This platform should allow sensor monitoring, light control, automation management, and future expansion.

The second objective is to implement night-related safety scenarios, especially the automation of night path lighting and night mode behavior. The goal is to support safe movement through the apartment during night hours by using appropriate light behavior and automation logic.

In addition to this, the project also aims to implement an adaptive night mode automation. The system keeps logs of when night mode is activated and deactivated over time. Based on these recorded patterns, the intention is to identify recurring habits and use them to switch night mode on and off more automatically in the future. This adds a more intelligent and personalized layer to the system, reducing the need for manual activation and making the solution more supportive of daily routines.

The third objective is to integrate and test relevant devices already present in the Experience Lab, including the Nuki smart lock, available lighting, sensors, and existing smart-home infrastructure, and connect them where possible into one coherent system.

The fourth objective is to implement routine-based automations that support everyday actions and reduce cognitive load. This includes logic for sleep-related modes, supportive lighting behavior, and emergency or helper-oriented actions where relevant.

The fifth objective is to integrate voice control through Alexa, so that certain routines and actions can be triggered by voice. At this stage, this part is still under development. The connection has already been explored through Emulated Hue.

The sixth objective is to deliver not only a working prototype, but also a complete documentation package for Mobilab & Care. This includes the project plan, the realization document, the user manual, and the installation process document. These documents should ensure that the project remains understandable, reproducible, and reusable after the internship.





Business Case

1. Situation after the internship assignment

After the internship, Mobilab & Care will have a working smart-home integration prototype in the Experience Lab that demonstrates how safety, comfort, and routine support can be combined in a realistic home environment. The system will be built around Home Assistant as a local integration platform and will connect lighting, sensors, automations, and selected existing devices in the apartment into one demonstrable setup.

The result of the internship will not only be the prototype itself, but also practical insight into how such a system can be configured, tested, and tuned in a living-lab context. In addition, the project will provide knowledge about technical limitations and dependencies, such as network restrictions, compatibility issues, and the effect of infrastructure constraints on reliability and maintenance.

2. Why the internship assignment is needed

Many smart-living and ageing-in-place concepts remain theoretical unless they are translated into a working implementation. This internship assignment is needed to explore how safety-oriented smart-home support can be realized in practice, using technologies that are realistic, demonstrable, and relevant for applied research.

The assignment is also needed because successful smart-home solutions depend on more than automation logic alone. They also require reliable device integration, clear user interaction, reproducible installation steps, and an understanding of practical barriers such as permissions, network dependencies, and interoperability. The internship therefore helps bridge the gap between concept and usable implementation.





3. Added value for the company

This project will provide Mobilab & Care with a reusable smart-home integration setup in the Experience Lab, built around a local control platform and supported by scenario-based automations. It will also provide practical knowledge about integrating devices such as smart lighting, sensors, the Nuki lock, and voice control through Alexa.

A further added value is that the project includes logging and documentation, making it possible to analyse system behavior and support future improvements. In particular, the project explores not only night-related automations, but also an adaptive night mode concept in which system logs can later be used to support more automatic activation and deactivation of night mode based on recurring patterns.

Finally, the project will produce a set of documents that remain useful after the internship: the project plan, the realization document, the user manual, and the installation process document. Together, these outputs support future demonstrations, reuse, and further development within Mobilab & Care.

4. Link between the assignment and business objectives

The project supports the broader mission of Mobilab & Care by contributing to the development and evaluation of practical smart-living solutions that are feasible, user-oriented, and relevant in care-related contexts. Because the work is carried out in the Experience Lab, it allows realistic testing instead of purely theoretical design.

The project also aligns with the group's interest in living-lab experimentation, reproducibility, and technology that can support independence and quality of life. By combining implementation, testing, and documentation, the assignment contributes both to technical development and to the creation of knowledge that can be reused in future research and demonstration activities.



Planning

Internship period: 23/02 - 22/05.

First phase: Understanding the project and feasibility (23/02 - 08/03)

This phase focuses on understanding the assignment, defining the scope, and validating the feasibility of the concept before implementation begins. The goal is to translate the main smart-home support scenarios into clear flows and determine which technologies and integrations are realistic within the Experience Lab context.

Feasibility is considered from several perspectives. The first is technical feasibility: whether the selected devices and platforms can communicate reliably and support the intended automations. The second is practical feasibility: whether the setup can be installed, configured, and maintained in a realistic way. The third is usability feasibility: whether the intended interactions remain simple, supportive, and appropriate for the target context.

By the end of this phase, the project direction is defined, the main use cases are selected, and the first version of the project plan is prepared, including scope, deliverables, and timing.

Second phase: Platform implementation (09/03 - 22/03)

The second phase focuses on building the technical foundation of the prototype. Home Assistant is prepared as the central local platform on Raspberry Pi, and the supporting system components are configured to allow integration, testing, and expansion. The dashboard environment is also prepared to support development and demonstration.

During this phase, the first devices are connected and the core system structure is established. This includes preparation for lighting control, sensor integration, automation logic, and monitoring. Logging is also considered from an early stage, so that relevant system events can later be captured and analysed.

In parallel, supporting technical documentation is prepared, including configuration notes and the first steps of the electrical and installation overview.



Third phase: Scenario implementation (23/03 - 26/04)

This phase focuses on developing the core functionality of the project. The main smart-home scenarios are translated into automations and integrated into the Experience Lab setup. Particular attention is given to night-related safety behavior, including night path lighting and night mode logic.

Within this phase, the project also includes the development of a night mode automation that not only controls behavior during night hours, but also records when night mode is activated and deactivated. These logs are intended to support future analysis of repeated patterns, with the goal of making night mode activation more adaptive and automatic over time.

Relevant devices already present in the Experience Lab, such as the Nuki lock and existing smart-home components, are incorporated where possible into a coherent setup. Voice-control functionality through Alexa is also explored in this phase, including the use of Emulated Hue as a possible integration path. In addition, helper-oriented actions, such as triggering calls or notifications through Alexa, are considered as part of the intended functionality.

This phase also includes tuning and adjustment of automations so that the scenarios are not only functional, but also suitable for demonstration and evaluation.

Fourth phase: Testing, tuning, and finalization (27/04 - 22/05)

The final phase focuses on validation, refinement, and documentation. The different automations and integrations are tested in the Experience Lab to assess reliability, consistency, and practical usability. Particular attention is given to the remaining open points, such as final Alexa validation in the lab environment, network-related dependencies, and the stability of the Nuki integration.

During this phase, the project is finalized as a demonstrable prototype and the documentation package is completed. This includes the project plan, the realization document, the user manual, and the installation process document. The goal is to ensure that the system is not only functional, but also understandable, reproducible, and maintainable after the internship ends.

Where possible, logs and test observations are used to support conclusions about behavior, limitations, and opportunities for future improvement.





Reporting and progress

During the internship, progress will be followed through weekly updates, technical testing, and ongoing documentation. The project will combine implementation work and documentation work in parallel, so that the final result is not only a functioning setup, but also a clear and reusable body of work for Mobilab & Care after the internship ends.

Technical progress will be documented through Home Assistant configurations, integration steps, automation logic, and practical testing in the Experience Lab. Challenges encountered during the project, such as hardware limitations, network issues, compatibility problems, and dependencies on IT permissions, will also be included in the reporting, since they are relevant to the feasibility of real-life implementation and future reuse.

The final reporting package of the internship will consist of:

- Project plan
- Realization document
- User manual
- Installation process document

Together, these deliverables will describe both the final implementation and the process followed to reach it. This includes installation choices, integration decisions, testing and tuning steps, limitations, and recommendations for future continuation or improvement.

